

SQL Query Practice – Revision Notes

Aggregate Functions, Nested Queries, and Execution Flow

1. Employee Table Schema

Column	Meaning
Eid	Employee ID
Ename	Employee Name
Dept	Department
Salary	Salary (e.g. 10k, 20k, 30k, 40k, 50k)

Common Aggregate Functions

Functions used to calculate summary values on a dataset:

MAX()	Maximum value
MIN()	Minimum value
AVG()	Average value
SUM()	Total sum of values
COUNT()	Number of rows

Query 1 – Find Maximum Salary

Question: Display the maximum salary from the Employee table.

```
SELECT MAX(Salary) FROM Employee;
```

Output: 50000

Query 2 – Find Employee Name Having Maximum Salary

Question: Display the name of the employee receiving the maximum salary.

⚠️ Why not `WHERE Salary = 50000`? Because we must dynamically query for the maximum salary without hardcoding values. We use a **subquery/nested query** instead.

```
SELECT Ename FROM Employee WHERE Salary = ( SELECT MAX(Salary) FROM Employee );
```

Output: Varun

4. Understanding the Nested Query Flow

Execution Flow Visualizer

OUTER QUERY: SELECT Ename FROM Employee
WHERE Salary =



INNER QUERY: SELECT MAX(Salary) FROM Employee



Inner Result Output: 50000



Evaluate Outer Condition: WHERE Salary = 50000



Final Output: Varun

Evaluation Table

Salary	Condition (Salary = 50000)
10000	✗ False
20000	✗ False
30000	✗ False
40000	✗ False
50000	✓ True (Returns 'Varun')